Home Lab Project Week 8, Day 3

Task:

Map the firewall interfaces to the appropriate VirtualBox networks and verify connectivity between the firewall, Kali Linux, and UTM.

Part 1 — Map the firewall interfaces to VirtualBox networks

This means giving OPNsense **two separate network adapters**, each attached to a distinct VirtualBox network type, so WAN and LAN are genuinely isolated from each other at the hypervisor level — not just logically inside OPNsense.

1. Shut down the OPNsense VM (must be off to change adapters).
2. VirtualBox → OPNsense → **Settings** → **Network**.
3. **Adapter 1 (→ will become WAN):** Attached to: **NAT** (or **Internal Network**, if you want a fully closed lab with no real internet access at all — NAT is simpler and fine for this lab).
4. **Adapter 2 (→ will become LAN):** Attached to: **Internal Network**, name it something consistent like `lab-lan` — and **this exact same network name must also be used on Kali's adapter.**
5. Go check **Kali's** Settings → Network → make sure its adapter is attached to **Internal Network**, same name (`lab-lan`). This is what puts Kali and OPNsense's LAN side on the same virtual wire.
6. Start OPNsense.

Part 2 — Assign WAN/LAN inside OPNsense (console, Option 1 then 2)

Log in (`root` / `opnsense`), then:

- Option **1** (Assign Interfaces): now that there are two adapters, you'll see two interface names (e.g., `vtnet0`, `vtnet1`). Assign one as WAN, the other as LAN, matching which VirtualBox adapter is which.
- Option **2** (Set Interface IP): give **LAN** a static IP in your lab subnet, e.g. `192.168.78.1/24` (matching Kali's existing `192.168.78.x` range). WAN can stay on DHCP from NAT.

Part 3 — Verify connectivity (the actual deliverable for this task)

From **Kali**, set its own adapter (if not already) to a static IP on the same subnet, e.g. `192.168.78.3/24`, then test:

ping 192.168.78.20

This should succeed — confirms Kali can reach OPNsense's LAN interface.

curl -k https://192.168.78.20

Confirms the web GUI is reachable over HTTPS (the `-k` ignores the self-signed cert warning).

For **Ubuntu**: since it's not yet wired into OPNsense's LAN or DMZ (that's Week 10's job), for now just confirm Kali can still reach Ubuntu directly as before:

ping 192.168.78.5

Screenshots to capture for Day 3:

- VirtualBox Network settings for OPNsense (showing both adapters and their types/names)
- VirtualBox Network settings for Kali (showing matching Internal Network name)
- The OPNsense console after Option 1/2, showing WAN/LAN assigned with IPs
- Kali's successful `ping 192.168.78.20`
- The `curl` or browser screenshot reaching the OPNsense web GUI